



Plant Microscope Lab



What are plants made of?

Plants are made of tiny parts called cells. Let's look inside them with the microscope!

TIME

60 minutes

GROUPS

Pairs at each microscope

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can use a microscope carefully, starting with the smallest lens.
- ✓ I can look closely and draw what I see.
- ✓ I can compare cells from different plant parts.

STEAM in this activity

S

SCIENCE

Plants are made of tiny cells.

T

TECHNOLOGY

A microscope makes small things look bigger.

E

ENGINEERING

Prepare a slide so light can pass through.

A

ART

Scientific drawing, and a cell-pattern artwork.

M

MATHS

Magnification: 4×, 10×, 40× bigger.



Before the lesson

- ☐ Check microscope lights and clean the lenses.
- ☐ Prepare leaf, stem, root and petal pieces.
An adult cuts thin slices.
- ☐ Onion skin and (optional, adult only) iodine for the stain test.
- ☐ Glass slides and cover slips; tissues for wiping.



Safety

- Ask an adult to cut the thin stem slice.
- Slides are glass. Hold them by the edges.
- An adult looks after the iodine. It stains skin and clothes.
- Wear gloves. Keep iodine away from your eyes and mouth.
- An adult helps with the nail varnish. Open a window – it has a strong smell.
- Wash your hands after.
- An adult cuts the onion.
- Onions can make your eyes sting. Do not rub them.
- An adult does all the cutting.
- Food colouring stains. Wear an apron.

Kit per group

- | | | |
|--|---|---|
| ☐  Microscope | ☐  Slides and cover slips | ☐  Dropper with water |
| ☐  Tweezers | ☐  A small leaf | ☐  A very thin slice of stem |
| ☐  A root (cress or bean sprout is best) | ☐  A petal | ☐  Two slides and cover slips |
| ☐  A piece of onion | ☐  Iodine solution (adult) | ☐  Gloves |
| ☐  A fresh leaf (not hairy) | ☐  Clear nail varnish (adult) | ☐  Clear sticky tape |
| ☐  Two slides | ☐  A red onion | ☐  Plain water |
| ☐  Very salty water | ☐  A celery stick with leaves | ☐  A glass of water |
| ☐  Food colouring (red or blue) | ☐  Knife (adult) | ☐  Microscope and a slide |

Running order (60 minutes)

0–10 min Story: what is a cell? Show the microscope parts.

10–30 min Plant parts tour.

30–50 min Onion and iodine tests.

50–60 min Share drawings and clear up the slides.

The investigations

Test 1: Plant Parts Tour: Which plant part will look the most green?

Children record: Which lens?; What colours can you see?; What shape are the cells?; Draw what you see

What scientists often see: Leaves are usually the greenest part. The green comes from tiny green blobs called chloroplasts. They use sunlight to make food for the plant. Roots grow in the dark, so they are usually not green.

Test 2: Onion Skin Cells: Will iodine stain make the onion cells easier to see?

Children record: Which lens?; How clear are the cells?; What shape are the cells?; Can you see a dark dot inside each cell?; What colours can you see?; Draw what you see

What scientists often see: Onion cells look like bricks in a wall. Iodine often turns them yellow-brown and makes a dark dot in each cell easier to see. That dot is the nucleus - the cell's control centre.

Test 3: Leaf Mouths: Which side of a leaf has more tiny mouths (stomata)?

Children record: Which lens?; Tap once for each tiny mouth; Draw what you see

What scientists often see: Many leaves have more stomata on the bottom side. It is shady and cool there, so the leaf loses less water. Some plants are different - water lilies have theirs on top!

Test 4: Salty Water Test: What will salty water do to red onion cells?

Children record: Which lens?; The purple part of the cells...; What colours can you see?; Draw what you see

What scientists often see: In salty water, water moves OUT of the cells. The purple part shrinks and pulls away from the cell wall. Put the slide back in plain water and the cells can fill up again!

Test 5: Celery Drinking Test: Where will coloured water go inside a celery stem?

Children record: Which lens?; Where is the colour?; What colours can you see?; Draw what you see

What scientists often see: The colour usually shows as little dots in a ring. Each dot is the end of a tube called xylem. Xylem tubes carry water from the roots all the way up to the leaves - like drinking straws.

Talk about it

- What shapes did you see?
- Did every plant part look the same?
- Why does a stain help us see?

What to look for

Starter: Focuses with the smallest lens first and draws what they see.

Explorer: Compares two plant parts and describes a difference in the cells.

Art corner

Cell pattern art. Use the cell shapes you saw to make a colourful pattern, like a mosaic.

Maths corner

Starter: Circle the lens that makes things look biggest: $4\times$ $10\times$ $40\times$.

Explorer: Count the cells you can see across the view with the $10\times$ lens: _____. With the $40\times$ lens: _____. Why is it fewer?

Real-world link

Scientists and doctors use microscopes every day, to study plants, germs and medicines.

Going further

Look at a leaf from a different plant, like grass or spinach. Are the cells the same?